Tutorial No : 5

Tutorial Title: Implement Stack and Queue using arrays.
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Program:

```
#include <stdio.h>
#define MAX 5
int stack[MAX];
int top = -1;

void stack_full()
{
   printf("Stack is full!\n");
}
void stack_empty()
{
   printf("Stack is empty!\n");
}

void push(int item)
{
   if (top == MAX - 1)
   {
      stack_full();
      return;
   }
   top++;
   stack[top] = item;
   printf("%d pushed into the stack.\n", item);
}

int pop()
{
   int item;
   if (top == -1)
   {
      stack_empty();
      return -1;
   }
   item = stack[top];
```

```
    top--;

    return item;
}

void display()
{
    int i;

    if (top == -1)
    {
        stack_empty();
        return;
    }

    printf("\nStack elements are:\n");

    for (i = top; i >= 0; i--)
    {
        printf("%d\n", stack[i]);
    }
    printf("TOP position = %d\n", top);
}
int main()
{
    int choice, item, deleted;

    while (1)
    {
        printf("\n----- STACK MENU -----\n");
        printf("1. PUSH\n");
        printf("2. POP\n");
        printf("3. DISPLAY\n");
        printf("4. EXIT\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch (choice)
        {
            case 1:
                printf("Enter element to push: ");
                scanf("%d", &item);
                push(item);
                break;
```

```
            case 2:
                deleted = pop();
                if (deleted != -1)
                {
                    printf("Popped element = %d\n", deleted);
                    printf("TOP position = %d\n", top);
                }
                break;

            case 3:
                display();
                break;

            case 4:
                printf("Program terminated.\n");
                return 0;

            default:
                printf("Invalid choice! Please try again.\n");
        }
    }
    return 0;
}
```

Output:

Output

```
----- STACK MENU -----
1. PUSH
2. POP
3. DISPLAY
4. EXIT
Enter your choice: 1
Enter element to push: 14
14 pushed into the stack.

----- STACK MENU -----
1. PUSH
2. POP
3. DISPLAY
4. EXIT
Enter your choice: 1
Enter element to push: 22
22 pushed into the stack.
```